Decoding the Qur'an's Muqaṭṭaʿāt through Abjad mathematics, Arabic linguistics and computational analysis

Exploratory research report: reproducible pipeline, pre-registered primary test, 10,000-trial randomized controls

Answer to the central question

The data do not support an encoded numerical-linguistic structure that performs better than chance. Under the pre-registered primary test, the 14 real openings decoded with every method together produce a score T = 36.0, against 32.3 ± 7.5 for random letter sequences drawn from the same 14-letter inventory (empirical p = 0.292, 10,000 trials). No single method survives correction for the eight methods tried (smallest Holm-adjusted p = 0.63). When the best method is chosen on half of the openings and tested on the other half, it beats chance in only 1.5% of 200 splits (median held-out p = 0.44). Across 660 opening-root pairs, none is enriched in its own surahs under all three robust tests after correction for multiple comparisons.

What the data do show: (1) every Abjad total in the brief is arithmetically correct; (2) the transformations generate real Arabic words easily, for random letters just as for the real openings (98.5% of random sequences yield at least one Qur'anic root, with 28.0 candidate roots on average versus 27.7 for the real openings); (3) a few candidate roots are genuinely more frequent in their surahs (علم in the Alif-Lām-Mīm surahs, أب in the Alif-Lām-Rā surahs, which is the Joseph narrative), but random letter sequences find comparably strong matches in the same surahs, so these are properties of the surahs' vocabulary, not of the letters.

The أب result is concentrated in Surah Yusuf: 28 of the 33 occurrences in the five Alif-Lām-Rā surahs are in surah 12, the story of Jacob and his sons.

Of the six preliminary discoveries, A, B and D are linguistically valid words but are **not** statistically enriched in their surahs; C (علم) is enriched but only as a subset match; E (ملك) is borderline and its equal-value twin كلم never occurs in Sad; F (طسم/طس/طسم) shows no measurable structural difference. Details follow.

1. Data, provenance and normalization

Primary corpus. Quranic Arabic Corpus (QAC) morphology v0.4 (Kais Dukes, built on the Tanzil Uthmani text 1.0.2), obtained verbatim from the GitHub mirror cltk/arabic_morphology_quranic-corpus because corpus.quran.com did not resolve from the build environment. File sha256 and every URL are listed in the workbook's Provenance sheet. The parsed corpus has 77,429 words, 49,967 root-bearing word tokens, 1,642 QAC roots, and 4,831 vocalized lemmas (4,341 unvocalized lemma spellings); 476 of the roots arise as candidates for some opening.

Cross-check. mustafa0x/quran-morphology (an Arabic-script revision of the QAC) agrees with the primary roots on 98.5% of words; the differences are later corrections (e.g. الناس assigned to نوس in v0.4 versus أنس in the revision) and roots added for proper nouns. All statistics use QAC v0.4.

General lexicon. Arramooz Alwaseet (T. Zerrouki; ~30k nouns, ~14k verbs, each with root; GPL) validates words that are Arabic but absent from the Qur'an. **Metadata:** Meccan/Medinan from quran.com's chapter data, cross-checked against fawazahmed0/quran-api; revelation order is the Egyptian standard order as distributed by Tanzil (transcribed because tanzil.net was unreachable; verified to be a permutation of 1-114).

Not obtained: Birzeit Quran Corpus, MASAQ (Mendeley) and corpus.quran.com were DNS-blocked from the execution environment, and no Buckwalter analyzer or classical dictionary (Lisān al-ʿArab, Lane) was available in machine-readable form. English glosses are analyst-supplied core meanings (flagged as such) and should be checked against Lane or Hans Wehr.

Normalization rules (applied identically to real and random sequences, and flagged on every candidate): all alif and hamza forms (ا أ إ آ ٱ ء) = ا = 1; ؤ = و = 6; ئ = ي = 10; ى = ي = 10; ة = ه = 5. Harakat and Qur'anic annotation marks are removed. An alif in a Muqaṭṭaʿāt inventory matching a hamza in a root is flagged as hamza normalization.

2. Phase 2: the 14 openings verified

opening	surahs	letters	Abjad	brief	verified	dig. root	place value	digits	dig.-root letter	partitions (<=4)
الم	2, 3, 29, 30, 31, 32	ا=1 + ل=30 + م=40	71	71	True	8	عا	زا	ح	41
المص	7	ا=1 + ل=30 + م=40 + ص=90	161	161	True	8	قسا	اوا	ح	30
الر	10, 11, 12, 14, 15	ا=1 + ل=30 + ر=200	231	231	True	6	رلا	بجا	و	14
المر	13	ا=1 + ل=30 + م=40 + ر=200	271	271	True	1	رعا	بزا	ا	11
كهيعص	19	ك=20 + ه=5 + ي=10 + ع=70 + ص=90	195	195	True	6	قصه	اطه	و	21
طه	20	ط=9 + ه=5	14	14	True	5	يد	اد	ه	40
طسم	26, 28	ط=9 + س=60 + م=40	109	109	True	1	قط	اط	ا	55
طس	27	ط=9 + س=60	69	69	True	6	سط	وط	و	40
يس	36	س=60 + ي=10	70	70	True	7	ع	ز	ز	47
ص	38	ص=90	90	90	True	9	ص	ط	ط	59
حم	40, 41, 43, 44, 45, 46	م=40 + ح=8	48	48	True	3	مح	دح	ج	34
حم عسق	42	م=40 + ع=70 + س=60 + ق=100 + ح=8	278	278	True	8	رعح	بزح	ح	12
ق	50	ق=100	100	100	True	1	ق	ا	ا	64
ن	68	ن=50	50	50	True	5	ن	ه	ه	37

All 14 totals in the brief are correct. Note that the brief's table covers 29 surahs; the seven Ḥā-Mīm surahs are 40-46, of which 42 adds ʿAyn-Sīn-Qāf and is analysed as its own opening.

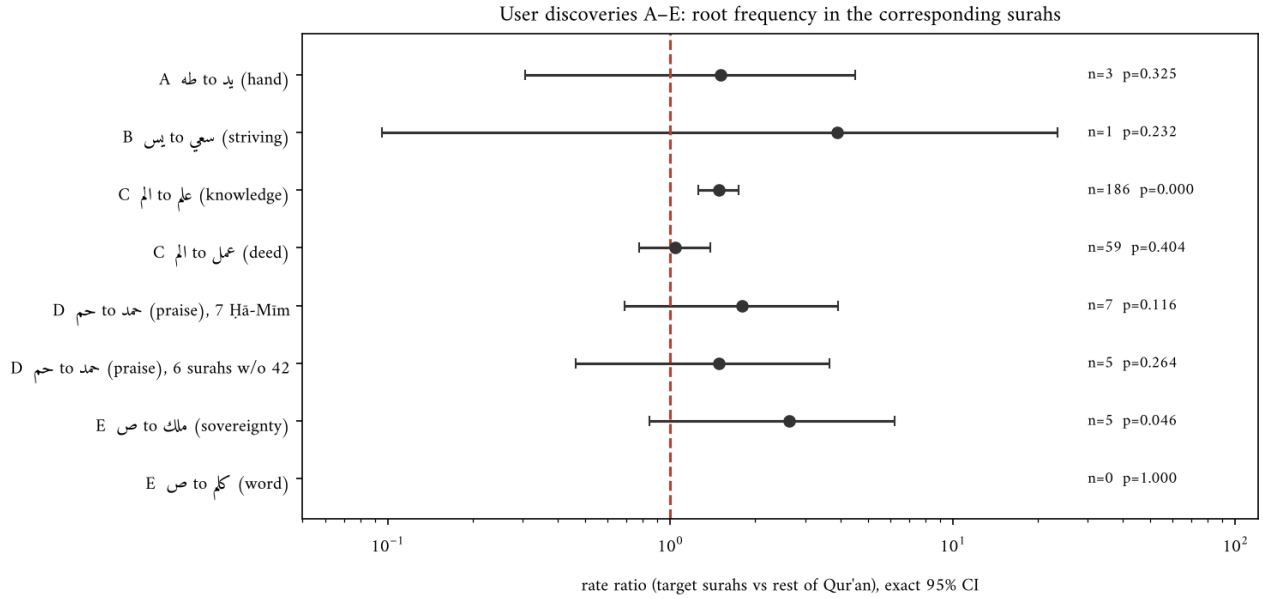
3. Methods

Decoding methods (Phase 4). M1 exact Abjad equivalence (every QAC root/lemma and Arramooz word with the same total); M2 decimal place-value decomposition (71 -> 70+1 -> عا); M3 individual digits, zeros dropped (71 -> 7,1 -> زا); M4 digital root (71 -> 8 -> ح); M5 original letters concatenated with a reconstruction, in either order; M6 all additive partitions of the total into at most four Abjad letters; M7 exact anagrams; M8A all original + reconstructed letters rearranged; M8B subsets of at least two letters containing at least one original and one reconstructed letter. M9 numerical relations between openings. Matching is by letter multiset for anagram methods and by exact spelling for ordered methods.

Frequency unit and denominator (Phase 6). The QAC root of each word stem; rates are per 1,000 root-bearing tokens. Each opening-root pair is compared with five baselines (all other surahs; other Muqaṭṭaʿāt surahs; surahs without Muqaṭṭaʿāt; surahs within ±35% in length; same Meccan/Medinan class within ±10 places of revelation order) using exact conditional-binomial rate ratios with Clopper-Pearson 95% CIs. Because word counts are bursty (over-dispersed), two surah-level tests are added: a negative-binomial GLM with a log-length offset (dispersion estimated from the intercept-only model, floored at 0.05; roots under 20 occurrences excluded), and a permutation test that resamples same-size surah sets 10,000 times. Benjamini-Hochberg FDR is applied within each test and globally across all 4,620 tests.

Randomized controls (Phase 7). Pre-registered before any statistic was computed (PREREGISTRATION.md, committed first). For each opening the score is the strongest enrichment (max -log10 hypergeometric p) among all its candidate roots; T sums this over the 14 openings. Taking the maximum means the random sequences enjoy the same freedom to pick their best match as the real ones, which is what controls the 'garden of forking paths'. Primary null N2: random sequences of the same length with distinct letters from the 14 Muqaṭṭaʿāt letters, same target surahs. Sensitivity: N1 (any of 28 letters), N3 (28 letters, Abjad total within ±15%), and LABEL (the real candidates scored against random surah sets of the same size). 10,000 trials each.

4. Phase 3: the preliminary discoveries, reproduced and tested



Rate of each root in the corresponding surahs relative to the rest of the Qur'an (exact 95% CI).

discovery	n	/1000 target	/1000 rest	RR	CI low	CI high	p exact	p NB	p perm	rank /1642
A طه → يد (hand)	3	3.59	2.38	1.51	0.306	4.51	0.325	0.31	0.201	195
B يس → سعي (striving)	1	2.28	0.586	3.9	0.0955	23.5	0.232	0.269	0.116	102
C الم → علم (knowledge)	186	23.5	15.9	1.48	1.25	1.75	2.88e-06	0.0123	0.0662	8
C الم → عمل (deed)	59	7.47	7.16	1.04	0.776	1.38	0.404	0.135	0.417	335
D حم → حمد (praise), 7 Ḥā-Mīm	7	2.15	1.2	1.79	0.688	3.94	0.116	0.201	0.19	97
D حم → حمد (praise), 6 surahs w/o 42	5	1.82	1.23	1.48	0.464	3.67	0.264	0.352	0.287	179
E ص → ملك (sovereignty)	5	10.7	4.06	2.63	0.845	6.24	0.0457	0.0556	0.106	53
E ص → كلم (word)	0	0	1.52	0	0	5.33	1	1	1	236

A. طه -> يد. Arithmetic verified: 9+5 = 14 = 10+4. The same total is produced three ways (M1 equal value, M2 place value, M6 partition). The lemma يد occurs in Ta-Ha at 20:22, 20:71, 20:110 (20:22 is the verse the brief cites). But the root ي-د-ي is not enriched: 3 occurrences, RR 1.51 (95% CI 0.31-4.51), p = 0.33, rank 195 of 1,642 roots in Ta-Ha. **Exceptionalism:** 8 Qur'anic lemmas and 6 roots have Abjad 14, plus 67 Arramooz words; one of them, the lemma أهواء (desires, Abjad 1+5+6+1+1 = 14), belongs to the root هوي, which is more enriched in Ta-Ha than يدي. What is unusual is narrower: M2 rarely yields a valid word at all, so the event 'place-value decoding gives a real word that occurs in the surah' has p = 0.022 against random sequences, uncorrected. Across the 126 opening x method tests this does not survive correction (q = 1.00). Verdict: valid word, exact equality, not statistically exceptional.

B. يس -> سعي. Arithmetic verified: 10+60 = 70 = ع; {ي س ع} rearranges to سعي (M8A, all letters used once); Abjad(سعي) = 140 = 2 x 70. Linguistically valid (lemmas سعى/سعي). But the root occurs **once** in Ya-Sin (36:20, the verse cited): RR 3.9 with a CI of 0.10-23.5, p = 0.23. A single occurrence cannot establish prominence. Verdict: valid, exact, not enriched.

C. الم -> علم / عمل. Correction to the brief: strict place-value decoding gives 71 -> 70 + 1 -> ع + ا, so the combined inventory is {ا ل م ع} and both words leave **two** alifs unused (one if the units digit is ignored). Classified as subset matches (M8B). علم **is enriched** in the six surahs: 186 occurrences, 23.5 vs 15.9 per 1,000 (RR 1.48, CI 1.25-1.75, exact p <0.0001; NB p 0.012; permutation p 0.066). It also holds against same-period and similar-length surahs. It is driven by al-Baqara (26.3/1000) and al-ʿAnkabut (36.7); as-Sajda is below the Qur'anic average. عمل **is not enriched** (RR 1.04, p 0.40). But random letters do as well in these surahs: the M8B score for الم is beaten by 16% of random sequences, because علم is a very frequent root in long surahs that any letter-rich inventory reaches.

D. حم -> حمد. Not an equal-value match: Abjad(حمد) = 52, not 48. Derivation: digits 4, 8 -> د ح; with the original م this gives {ح م د} with one ح unused (M8B). In the seven Ḥā-Mīm surahs the root occurs 7 times (RR 1.79, CI 0.69-3.94, p 0.12); without surah 42, 5 times (p 0.26). Verdict: not enriched. The strongest M8B root for حم is حمم (hot water / close friend), not حمد.

E. ص -> ملك / كلم. Both have Abjad 90 (M1, M6). ملك occurs 5 times in Sad (RR 2.63, CI 0.85-6.24, exact p 0.046, NB p 0.056, permutation p 0.106): nominally borderline, not significant after correction. كلم occurs **zero** times in Sad. Nine roots share the value 90; ملك is the only one occurring in Sad. Against random sequences the M1 score for ص is p = 0.36. Verdict: thematically apt but statistically weak.

F. طسم (109) -> طس (69) -> طسم (109). The difference is 40 = م, verified. Lexically, surah 27 is not more different from its neighbours than they are from each other: Jensen-Shannon divergence of root distributions 26-27 = 0.364, 27-28 = 0.284, 26-28 = 0.351; the middle-distinctiveness statistic is at the 40th percentile of the 25 adjacent surah triples with at least 500 tokens each. The letter م is not depleted in 27 (83.5 per 1,000 letters vs 86.4 in 26, 78.1 in 28, 82.1 Qur'an-wide). Its distinctive roots (مكر plot, طير birds, عرش throne, نمل ants) reflect the Solomon and Saba narrative; none survives FDR correction. Verdict: no measurable structural correlate of the missing م.

5. Phases 4-5: exhaustive generation and validation

The nine methods produced 3,551 candidate records (2,773 unique opening-method-word combinations), validated as follows (2,198 unique words):

validation status	unique candidates
Arabic word (Arramooz), root not in the Qur'an	816
exact Qur'anic lemma	650
Arabic word (Arramooz) whose root occurs in the Qur'an	555
exact Qur'anic surface form	94
Qur'anic root (bare root letters)	83

Phase 7 question 1 and 2: how often do random sequences produce legitimate words, and do they occur in the assigned surahs?

sequences	>=1 valid Qur'anic root	mean candidate roots	best root occurs in target surah	best root p<0.001
N2	0.985	28	0.934	0.259
N1	0.982	17.9	0.875	0.158
N3	1	29	1	0.215
REAL	1	27.7	1	0.286

Random letters produce valid Qur'anic roots almost as reliably as the real openings, and their best root occurs in the target surahs most of the time. Producing a real word, and finding it in the surah, is therefore the expected outcome of these transformations, not evidence of design.

6. Phase 6: cross-referencing all 114 surahs

660 opening-root pairs were tested; 193 roots occur at least once in their target surahs, 28 are nominally enriched against the rest of the Qur'an ($p < 0.05$), 7 under the over-dispersion-aware NB GLM, and 27 under the surah permutation test. At about 5% false positives, 33 nominal hits are expected by chance alone. After global FDR correction:

test	pairs with q<0.05	which
B1_all_other_surahs	7	مدن:طسم, قوم:المص, قول:المص, نفس:الم, علم:الم, جيا:الر, ابو:الر
B2_other_muqattaat_surahs	4	قوم:المص, نفس:الم, علم:الم, ابو:الر
B3_surahs_without_muqattaat	7	مدن:طسم, قوم:المص, قول:المص, علم:الم, ربب:الر, جيا:الر, ابو:الر
B4_similar_length_(±35%)	6	قوم:المص, قول:المص, نفس:الم, علم:الم, جيا:الر, ابو:الر
B5_same_period_(±10_in_revelation_order)	4	قوم:المص, نفس:الم, علم:الم, ابو:الر
NB_GLM	0	none
surah_permutation	0	none

No pair passes all three robust tests (token-level vs rest, NB GLM, permutation): 0 of 660. The token-level survivors (مدن, قول, قوم, جيا, أب, نفس, علم) are frequent words in long narrative surahs; the token-level test ignores burstiness, which is why they lose significance at the surah level.

opening	root	meaning	methods	n	RR	p vs rest	NB RR	NB p	perm p	p same period
الم	نفس	soul, self	M1	72	1.7	0.000124	1.84	0.00527	0.0289	0.000181
الم	علم	knowledge, know; sign; world	M8A, M8B	186	1.48	2.88e-06	1.47	0.0123	0.0656	2.22e-05

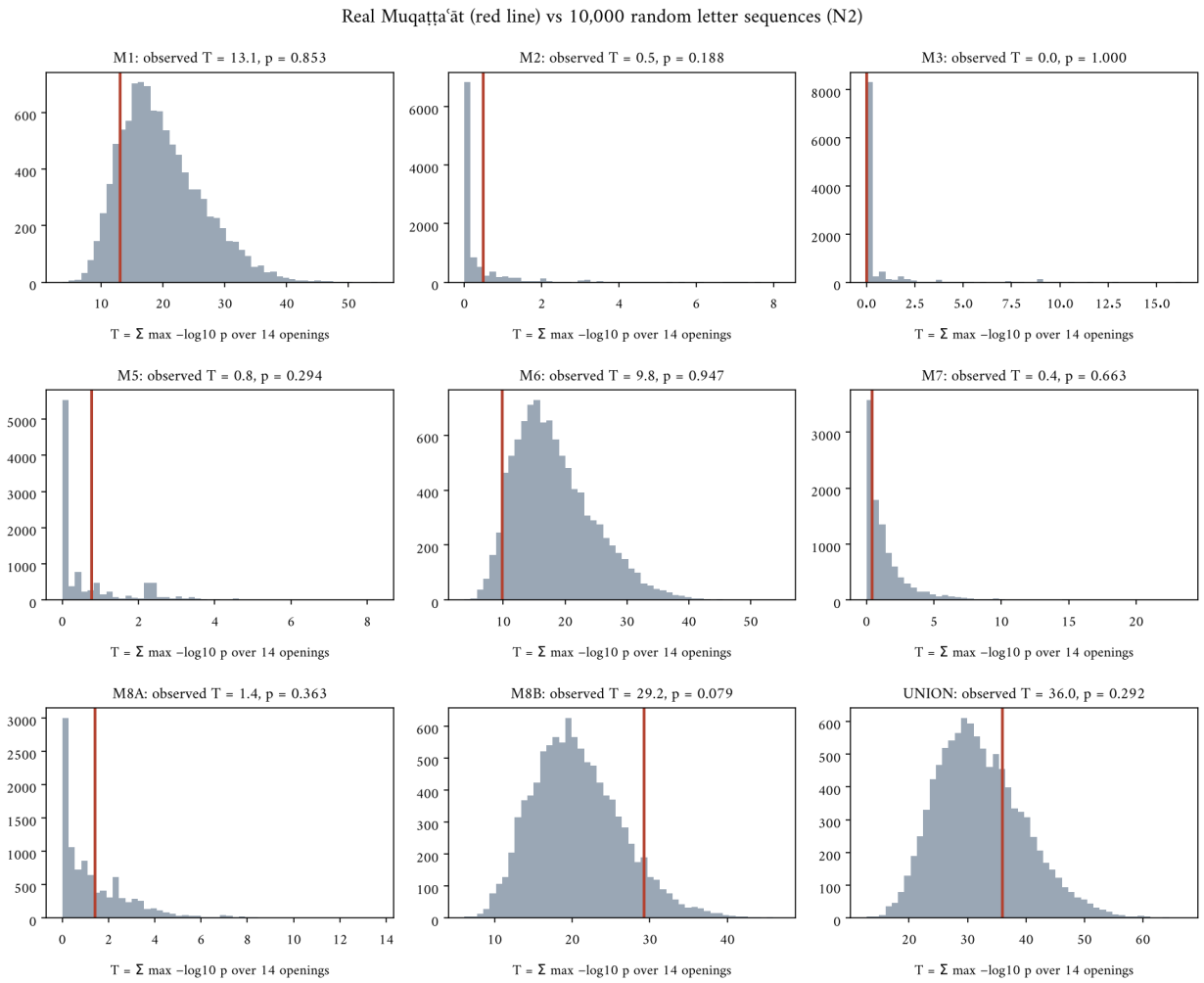
opening	root	meaning	methods	n	RR	p vs rest	NB RR	NB p	perm p	p same period
الر	جيا	come, bring	M8B	46	2.06	2.38e-05	1.94	0.013	0.0353	0.00155
الر	ابو	father	M8B	33	4.09	1.07e-09	3.46	0.0148	0.0421	5.99e-06
كهيعص	عصي	disobey	M8B	2	5	0.0661	4.99	0.0177	0.0864	0.0772
المر	عمي	blindness	M8B	2	5.66	0.0532	5.61	0.022	0.0294	0.0333
حم	جدل	dispute, argue	M1	6	4.49	0.00436	3.75	0.0383	0.0311	0.0406
الر	قطع	cut, sever	M1	6	2.08	0.0911	2.11	0.0552	0.142	0.0422
ص	ملك	king, sovereignty, possess; angel	M1, M6	5	2.63	0.0457	2.67	0.0556	0.105	0.0365
المر	موه	water	M8B	3	4.39	0.0344	3.92	0.0573	0.114	0.0137
الر	صصۤ	narrate stories; follow traces	M1, M6	6	2.6	0.0432	2.78	0.0628	0.105	0.096
المص	صصۤ	narrate stories; follow traces	M8B	5	4.46	0.00845	5.05	0.0656	0.0357	0.0551

7. Phase 7: randomized control experiment

method	T obs	T null mean	z (N2)	N2 p	Holm p (N2)	N1 p	N3 p	LABEL p
M1	13.1	19.8	-1.01	0.853	1	0.36	0.794	0.458
M2	0.488	0.321	0.223	0.188	1	0.195	0.433	0.162
M3	0	0.409	-0.302	1	1	1	1	1
M5	0.772	0.663	0.104	0.294	1	0.171	0.224	0.145
M6	9.83	18.3	-1.31	0.947	1	0.505	0.947	0.693
M7	0.417	1.44	-0.525	0.663	1	0.414	0.732	0.701
M8A	1.41	1.39	0.00707	0.363	1	0.179	0.316	0.246
M8B	29.2	20.7	1.48	0.0793	0.634	0.0105	0.107	0.0828
UNION	36	32.3	0.492	0.292	n/a	0.0425	0.165	0.1

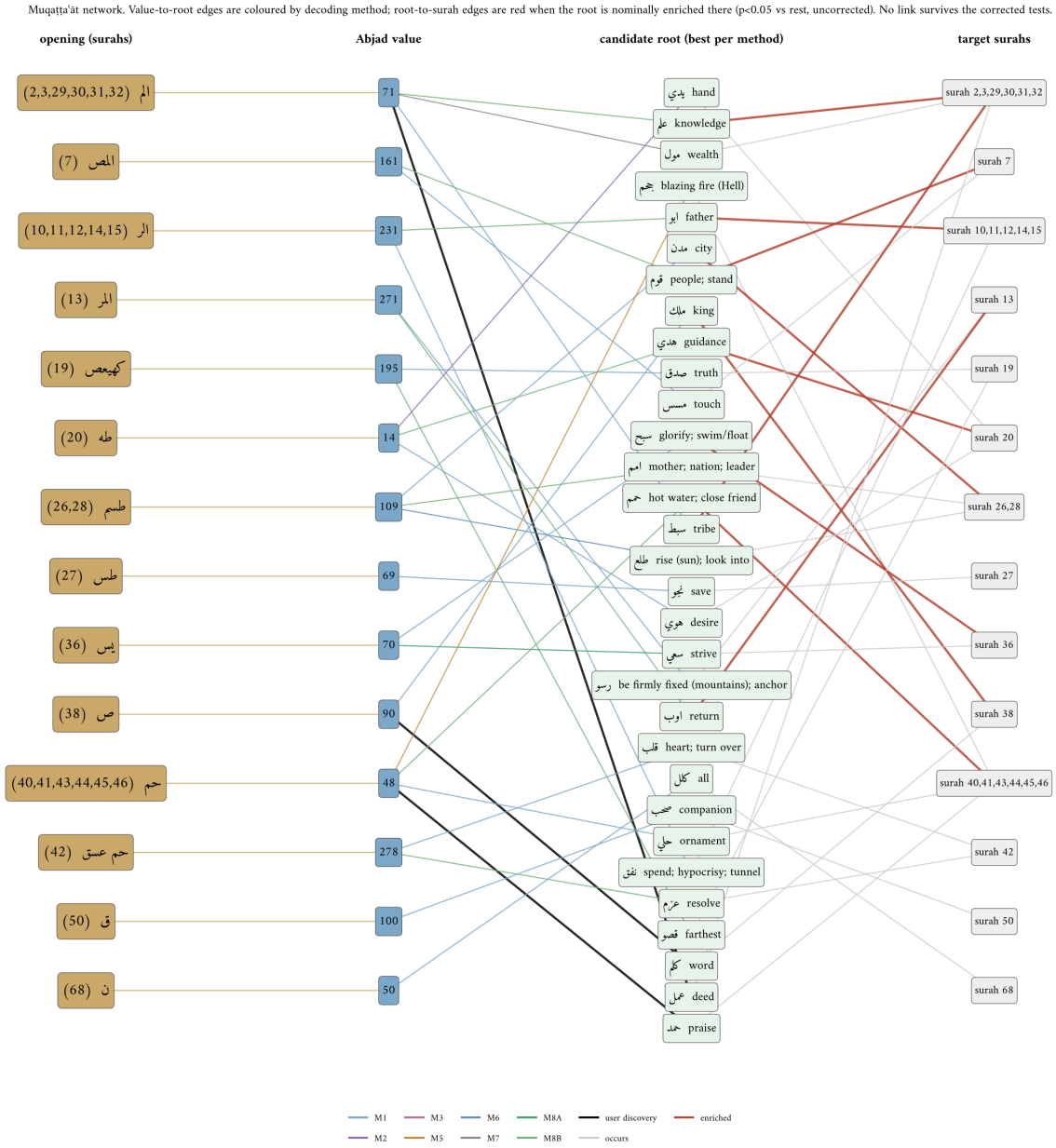
Primary (pre-registered) result: UNION vs N2, p = 0.292. H1 is not supported. The single sensitivity null in which the real openings look better is N1 (uniform over all 28 letters; UNION p = 0.042, M8B p = 0.010). N1 is the least appropriate comparison: the 14 Muqaṭṭaʿāt letters are frequent, light letters that form many more roots than rare letters such as ظ, غ or ث (18 candidate roots per random N1 sequence vs 28 for N2). Once the letter inventory (N2) or the Abjad magnitude (N3, p = 0.165) is matched, the advantage disappears. Specificity (LABEL): the real candidate sets are not better matched to their own surahs than to random surah sets of the same size (p = 0.100).

Held-out method selection. In 200 random 7/7 splits, the method chosen on the training openings was M8B (113), M2 (38), M8A (22), M5 (18), M7 (3), M1 (3), M3 (3); on the held-out openings the median p was 0.44 and only 3 of 200 splits reached $p < 0.05$, fewer than the 10 expected by chance. No method generalizes from one half of the openings to the other.



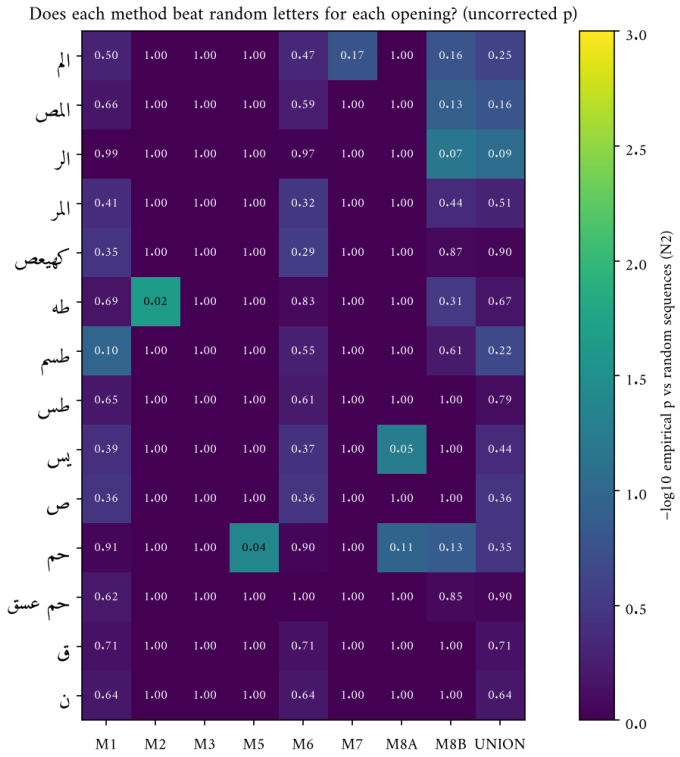
Distribution of T for 10,000 random letter sets (N2) with the real Muqaṭṭaʿāt marked in red.

8. Phase 8: the system as a network



Opening -> Abjad value -> best candidate root per method (and the user's discoveries) -> target surahs.

A genuine code should use a few rules consistently. Per opening and method, only 2 of 112 combinations beat random sequences at uncorrected $p < 0.05$ (طه M2 -> يدي ($p=0.022$); حم M5 -> جحم ($p=0.042$)), against about 6 expected by chance, each with a different method and none surviving FDR. No method works for more than one opening, and the user's own discoveries each require a different transformation (A: M2/M1, B: M8A, C: M8B with letters discarded, D: M3 + M8B with an unequal value, E: M1). That is the signature of post-hoc selection rather than of a rule.



Empirical p of each opening x method against random sequences (N2), uncorrected.

M9: numerical relations between openings. Relations among the 14 totals (equal digital roots, differences that are letter values, integer multiples) occur no more often than among random openings of the same lengths:

relation	observed	random mean	random sd	p (>= observed)
equal total	0	0.762	0.899	1
equal digital root	10	10.2	3.02	0.525
difference is a letter value	17	16.5	5.11	0.444
integer multiple	2	9.41	6.31	0.953
total is itself a letter value	4	4.3	1.02	0.772

Examples that look striking in isolation: Abjad(طه) x 5 = Abjad(يس); Abjad(ق) = 2 x Abjad(ن); الر and المر differ by 40 = م (as do طسم and طس); four single-letter openings (يس=70=ع, ص, ق, ن) have totals that are themselves letters. Random sets produce as many such coincidences, so they are not reported as findings.

9. Ranked findings

Each finding (best candidate root per opening and method, where it occurs in the target surah) is rated on six separate dimensions: D1 mathematical exactness (3 = all letters / equal value, 2 = subset); D2 linguistic validity (3 = Qur'anic lemma, 2 = root only); D3 rule consistency (number of openings, out of 14, for which that method beats random letters); D4 direct textual correspondence (occurrences in the target surahs); D5 statistical significance (global FDR q, NB GLM); D6 performance against random sequences (empirical p). No weights are used: findings are ranked by Pareto front, so a finding is ranked below another only if it is no better on every dimension and worse on at least one. The full ranking is in the workbook.

front	opening	method	root	meaning	D1	D2	D3	D4 n	D5 q	D6 p
1	طه	M2	يدي	hand	3	3	1	3	1	0.0223
1	الم	M8B	علم	knowledge, know; sign; world	2	3	0	186	0.757	0.163
1	الم	M7	مول	wealth	3	3	0	15	1	0.167
2	حم	M5	جحم	blazing fire (Hell)	3	2	1	3	1	0.0425
2	الر	M8B	ابو	father	2	3	0	33	0.826	0.0656
2	طسم	M1	مدن	city	3	3	0	5	n/a	0.104
2	المص	M8B	قوم	people; stand, establish	2	3	0	55	1	0.125
2	ص	M1	ملك	king, sovereignty, possess; angel	3	3	0	5	1	0.356
3	طه	M8B	هدي	guidance, guide; offering	2	3	0	10	1	0.314
3	كهيعص	M1	صدق	truth, sincerity; charity	3	3	0	4	1	0.352
3	المص	M1	مسس	touch	3	3	0	4	1	0.662
4	يس	M1	سبح	glorify; swim/float	3	3	0	3	1	0.39
4	طسم	M8B	امم	mother; nation; leader	2	3	0	8	1	0.607
5	حم	M8B	حمم	hot water; close friend	2	2	0	5	1	0.134
5	الم	M1	سبط	tribe	3	2	0	3	n/a	0.499
5	طسم	M6	طلع	rise (sun); look into	3	3	0	2	n/a	0.552
5	طس	M1	نجو	save, deliver; confide	3	3	0	2	1	0.646
5	طه	M1	هوي	desire, whim; fall	3	3	0	2	1	0.694
6	يس	M8A	سعي	strive, hasten, endeavour	3	3	0	1	1	0.0549
6	المر	M1	رسو	be firmly fixed (mountains); anchor	3	3	0	1	n/a	0.415
6	المر	M8B	اوب	return, turning back	2	3	0	2	n/a	0.437
6	حم عسق	M1	قلب	heart; turn over	3	3	0	1	1	0.615

0 findings pass all statistical bars (q < 0.05 for NB GLM, permutation and the random-sequence comparison). The strongest single finding on the mathematical and linguistic dimensions is طه -> يد (exact, Qur'anic lemma, present in 20:22, 20:71, 20:110), which fails on frequency. The strongest on frequency are علم (ALM surahs) and أب (ALR surahs), which are subset matches that random letters also reach.

10. Phase 10: the proposed thematic sentence

word	opening	how derived	n in target	RR vs rest	p vs rest	NB p	method vs random letters
علم knowledge	الم	M8A, M8B	186	1.48	2.88e-06	0.0123	M8B: p=0.163, q=1.00
عمل action	الم	M8A, M8B	59	1.04	0.404	0.135	M8B: p=0.163, q=1.00
سعي striving	يس	M8A	1	3.9	0.232	0.269	M8A: p=0.055, q=1.00
حمد praise	حم	M8B	5	1.48	0.264	0.352	M8B: p=0.134, q=1.00
ملك sovereignty	ص	M1, M6	5	2.63	0.0457	0.0556	M1: p=0.356, q=1.00

The five words come from three different transformations (M8B subset with discarded letters, M8A anagram, M1 equal value), one of them with an unequal value (حمد). Only علم is more frequent in its surahs, and the random controls show that is a property of al-Baqara and al-ʿAnkabut, not of the letters. Applying any single one of these rules uniformly to all 14 openings produces mostly unrelated words (e.g. under M8B: قوم people for المص, أب father for الر, أوب return for المر, قصو farthest for كهيعص, عزم resolve for حم عسق). The sentence is therefore a meaningful human synthesis of Qur'anic themes, but it is not generated by a consistent mathematical rule, and the data do not support it as an encoded message.

11. Failed hypotheses

- H1 (pre-registered): the Muqaṭṭaʿāt beat random letter sequences under the full decoding toolkit (p = 0.29).

- Any single decoding method works across openings (best Holm-adjusted p 0.63; held-out validation fails).

- Candidate words are specific to their own surahs (LABEL null p 0.10).

- Numerical relations among the openings exceed chance (all M9 p >= 0.44).

- Discoveries A (يد), B (سعي), D (حمد), the عمل half of C, and كلم in E are enriched in their surahs.

- Surah 27 is structurally or lexically distinct from 26 and 28 in a way linked to the missing م (F).

12. Limitations

- Root is the frequency unit. Theme was operationalized as lexical enrichment; human judgements of theme were not scored, because they cannot be applied blind to random controls.

- The QAC v0.4 is the 2011 release; later root corrections (~1.5% of words) would change some counts slightly.

- Birzeit, MASAQ, a Buckwalter analyzer and classical dictionaries could not be downloaded; the lexicon is the QAC plus Arramooz, a modern lexicon. Some classical words may be missed, though this affects real and random sequences equally.

- Abjad conventions (hamza, ة, ى) follow one stated scheme; others (e.g. ة = 400) would change some totals.

- Statistical power is limited for rare roots in short surahs: a word occurring once (سعي in Ya-Sin) can never be shown to be enriched, whatever its importance to the narrative.

- The search space was fixed in advance with explicit letter limits; a looser search (longer partitions, more transformations) would find more words for random letters too and cannot rescue H1.

Deviations from pre-registration. (1) The negative-binomial GLM originally estimated dispersion jointly and produced degenerate p ~ 0 for rare roots (e.g. منن, one occurrence in Sad). It was

replaced, after inspection, by a fixed-dispersion GLM with a 20-occurrence minimum. This affects only secondary Phase 6 tests and made them more conservative. (2) The rarefaction sample for Discovery F was reduced from 1,000 to 500 tokens because surahs 26-28 have fewer than 1,000 root tokens. The primary test and all null models are unchanged.

13. Conclusions

The Muqaṭṭaʿāt letters can be turned into meaningful Arabic words by Abjad arithmetic, and some of those words fit the themes of their surahs strikingly well (يد in Ta-Ha, سعي in Ya-Sin, ملك in Sad). Rigorous testing shows that this is what the transformations do to almost any short sequence of common Arabic letters: with nine methods, roughly 28 candidate roots per opening and long, lexically rich target surahs, good-looking matches are expected. The real openings do not outperform random letters, no rule works consistently, and no candidate is robustly enriched after correction. Within the scope of these methods and data, the evidence favours coincidence and selection over a reproducible numerical-linguistic code. This is a statement about these Abjad decoding procedures only; it says nothing about the Muqaṭṭaʿāt's meaning as understood in the tafsir tradition.

14. Reproducing every result

pip install -r requirements.txt; python scripts/run_analysis.py (downloads data, verifies sha256, runs Phases 2-8 with 10,000 trials, about 5 minutes on 4 cores); python scripts/build_outputs.py (workbook, figures, network); python scripts/build_report.py (this PDF). All random seeds are fixed. See README.md.